EQUALITY, EFFICIENCY AND THE FUTURE OF AMERICA

by

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Submitted to Princeton University
Department of Economics
In Partial Fulfillment of the Requirement for the A.B. Degree

April 3, 1997

"When I was young, it was common to hear my elders say that it was better to be poor in the U.S. than to be poor in other countries. But that's no longer true. The lowest 10% of wage earners in Italy, once the epitome of poor European countries, now makes three times more than their counterparts in the United States"

- U.S. Rep. George E. Brown Jr.

"The economic trends are forcing the society apart. The schools are not closing the gap. It'll widen. It's hard to run a democracy in which that's the prevalent expectation. You get anger and bitterness, and the wealthy distancing themselves - gated communities, private schools, private security forces. You have a Latin American country."

- Arnold Packer, Institute for Policy Studies

"Should a nation choose to push against the economic forces that would otherwise divide it, it has the ability to do so. The consequence of choosing otherwise - of pretending the choice is not ours to make - is eventually to cease being a society."

-Robert Reich, former Secretary of Labor to my mother and father

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Introduction

This thesis will study the relationship between equality and efficiency, or more precisely, the relationship between a county's income distribution and its aggregate productivity. The fundamental hypothesis is that the marginal effect of economic inequality is to adversely affect aggregate productivity in circumstances of extreme inequality and to promote aggregate productivity in circumstances of moderate inequality.

Chapter One presents a theoretical analysis of the tradeoff between equality and efficiency. It discusses the existing canon of economic research on the subject and synthesizes the two major opposing theories into a new complex relationship called the Morfit curve. Chapter Two provides empirical support for the theories of Chapter One. Linear regression analysis is used to demonstrate that empirical data square with the Morfit curve. Chapter Three examines the implications of the Morfit curve for the U.S. economy. It explains the present state of economic equality in the U.S., makes prognostications for the future and analyses the tradeoff between equality and efficiency which presents itself to policy makers in the U.S. today.

Chapter One

The Tradeoff: Equality vs. Efficiency

A Theoretical Basis

I. THESIS

Arthur Okun, the world-famous economist who made his career by empirically quantifying the tradeoff between unemployment and inflation, writes of the tradeoff between equality and efficiency that, "It is, in my view, our biggest socioeconomic tradeoff, and it plagues us in dozens of dimensions of social policy."[1]

In his book Equality and Efficiency: the Big Tradeoff, Okun presents the theoretical basis of the relationship between equality and efficiency. He writes that the most fundamental form of equality is a politically guaranteed right. Distributed universally, equally and free of charge, rights may not be bought nor sold, nor may they be used as incentives for certain behaviors. Their existence represents an encroachment upon individual liberty. Equal rights are barriers to the perfectly competitive market and are therefore sources of inefficiency.

If a society's set of rights does not include economic equality, and it organizes its economic activities through a market, inequalities of income are inevitable. These inequalities are due to a combination differences in natural intelligence, family circumstances and levels of education as well as random chance. The existence of such inequality is a significant impetus for efficiency and growth. Inequality is a motivating force - both a carrot and a stick. The hope of attaining riches and the fear of falling into poverty inspire men to work hard and develop their skills.

Okun states that the perfectly competitive market represents the extreme of productive efficiency. The inequalities that arise under perfect

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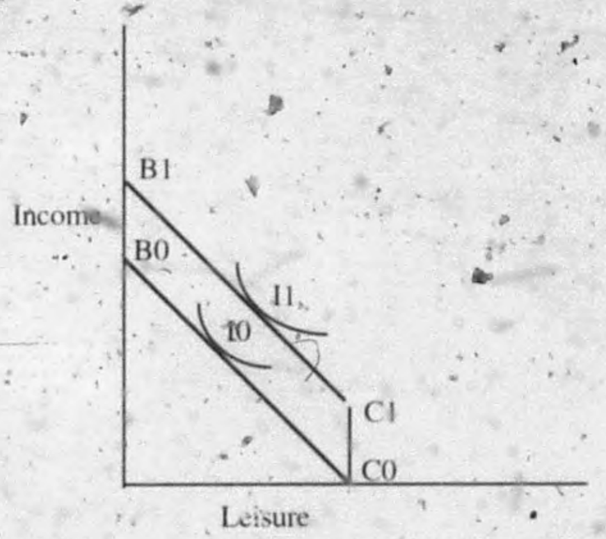
[1]Arthur Okun, Equality and Efficiency: the Big Tradeoff (Washington, DC: The Brookings Institution, 1975) p. 2.

competition serve as efficiency-maximizing incentives. Any government intervention in the economy which serves to redistribute and equalize income represents a step away from this efficiency maximum. In addition, the bureaucratic costs of administering such intervention create significant inefficiencies.

Social safety nets, which provide income regardless of effort, distort the behavior of laborers. The income effect of such programs drives men from laboring into idleness. The income effect of a cash transfer on labor supply is represented in Figure 1.1.

Figure 1.1

Income Effect of Transfer Payments on Labor Supply



The person's income is measured on the Y axis and his / her leisure hours are measured on the X axis. The line segment B_0C_0 is the person's budget constraint and its slope is determined by his / her hourly wage rate. The person's relative valuation of income and leisure is represented by the indifference curve I_0. The point of tangency between B_0C_0 and I_0 marks the amount of income and leisure that the person chooses. A government transfer increases the person's income by a lump sum and changes his / her budget constraint to $B_1C_1C_0$. This new budget constraint allows the person to reach indifference curve I_1. The amount of leisure and income chosen is now at the point of tangency of $B_1C_1D_1$ and I_1. This point represents an increase in leisure hours, and thus a decrease in labor hours, over the no-transfer situation.[2]

The income effect has an particularly perverse effect on workers whose wages are extremely low. The size of cash transfers varies inversely with wage earnings. When the worker's wages are so low that the income gained by increasing labor hours is largely offset by the income lost in reduced transfer payments, the worker will realize that his income is relatively constant regardless of how much he chooses to work. He will opt not to work at all and to be paid by the government to do nothing.

Social programs also distort patterns of savings and investment. The perfectly competitive market directs the funds of savers to the most productive investment loci: firms with the highest potential for growth and the highest gains for the investor. Government programs necessitate additional taxes, which the wealthy must incorporate into their savings behavior. This represents a distortion of savings behavior away from its optimality, and is therefore an inefficiency.

Okun employs the metaphor of a "leaky bucket" which carries money from the rich to the poor. "The leak represents an inefficiency... of real world

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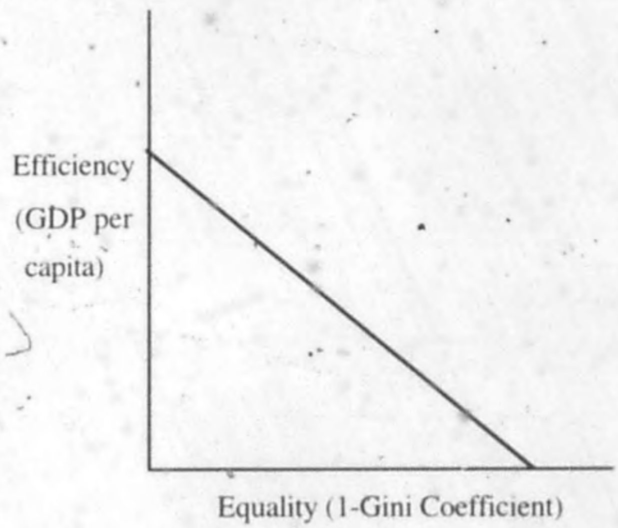
[2] C.V. Brown, Taxation and the Incentive to Work, (New York: Oxford University Press, 1980) p. 3.

redistribution. The adverse effects on the economic incentives of the rich and the poor, and the administrative costs of tax-collection and transfer programs."[3] Okun states that the relationship between equality and efficiency is a simple negative correlation. More equality must be bought at the price of less efficiency and vice versa.

Figure 1.2 is a graphical representation of the equality / efficiency tradeoff as conceived of by Okun. Efficiency is measured on the Y axis by GDP per capita and equality is measured on the X axis by a variable defined as the Gini Coefficient[4] subtracted from 1.

Figure 1.2

Negative Correlation (Okun)



[3]Okun p. 92.
[4]For an explanation of the Gini Coefficient see Appendix to Chapter One

Refer to figure 1.2, Okun's negative relationship. Since the Gini coefficient has a range of zero to one, equality defined as the variable (1 - Gini coefficient) also has a range of zero to one. An equality value of zero represents perfect inequality (all income flowing to one family). An equality value of one represents perfect equity of income. Generally speaking, there is no upper limit on GDP per capita, it can be perpetually augmented by increasing national resource endowment (technological innovation and education). However, in this context the efficiency measure is finite. By graphing equality versus efficiency we are implicitly assuming that all other factors which influence efficiency are held constant. Within these constraints, there is a maximum value of efficiency, which can be obtained by establishing a specific income distribution. It should also be noted that the slope of the line on figure 1.1 is arbitrary and is not a quantitative measure of the tradeoff between equality and efficiency. The graph merely represents the theoretical concept of a negative relationship.

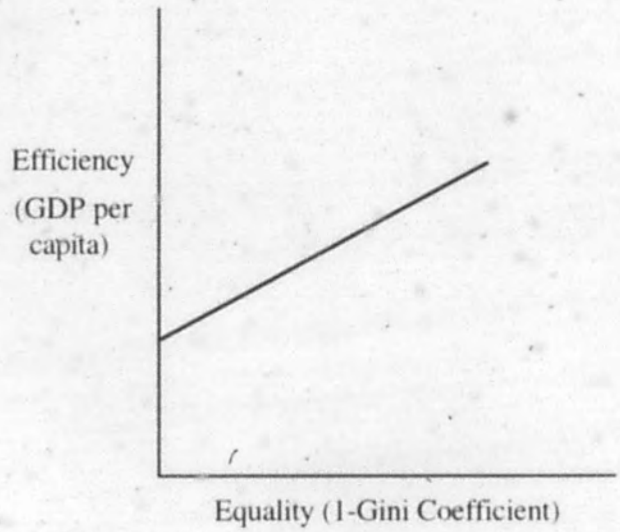
II. ANTITHESIS

Empirical Evidence for a Positive Correlation

Although economic theory seems to predict a negative correlation between equality and efficiency, several empirical studies have discovered just the opposite. In 1975, Jan Tinbergen performed a cross-sectional international analysis of statistics on income distribution and productive efficiency. Tinbergen's regression of GNP per capita versus inequality arrived at a regression coefficient of -0.019. Tinbergen's colleague Lydall found the coefficient to be -0.031. Since a negative effect of inequality on growth necessarily implies a positive effect of equality on growth Tinbergen and Lydall's findings supports the theory of a positive relationship between equality and efficiency (Figure 1.3).

Figure 1.3

A Positive Relationship



A Theoretical Explanation

A common economists' explanation for the positive correlation between equality and efficiency is best articulated and developed by sociological texts. Peter and Judith Blau, sociologists at the State University of New York at Albany write that

> socioeconomic inequalities undermine the social integration of a community... it creates a situation characterized by much social

> disorganization and prevalent latent animosities... There is much resentment, frustration, hopelessness and alienation.[5]

Indicators of social malaise, such as rates of divorce and migration, are linked to economic inequality. Rates of violent crime are particularly strongly correlated with inequality.

> The answer... is unequivocal: income inequality in a metropolis substantially raises its rate of criminal violence... This finding cannot be dismissed as indicating an indirect influence of poverty... apparently the relative deprivation produced by much inequality rather than the absolute deprivation produced by much poverty provides the most fertile soil for criminal violence.[6]

The Blaus' theory supports the findings of Tinbergen and Lydall. If economic inequality can causes significant social stress, then it can be detrimental to efficiency. An alienated working class which resents its bosses and hates its jobs is not a productive working class. An overly stressed society is a politically unstable one - a climate which discourages long-term investment and business enterprises. Generally speaking, a society with pervasive psychological discontent is not an economically efficient society.

More Empirical Evidence for a Positive Correlation

Twenty years after Tinbergen and Lydall found a positive correlation between equality and efficiency, Torsten Persson and Guido Tabellini asserted that inequality hinders growth in an article for the *American Economic Review*. They write that "Both historical panel data and postwar cross sections indicate a significant and large negative correlation between inequality and growth."[7] Persson and Tabellini used data from Austria,

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[5]Judith and Peter Blau, *The Cost of Inequality: Metropolitan Structure and Violent Crime*
[6]Blau
[7]Torsten Persson and Guido Tabellini, *Is Inequality Harmful for Growth?* American Economic Review June 1994, p. 600.

Denmark, Finland, Germany, the Netherlands, Norway, Sweden, the United Kingdom and the United States, from 1830 to 1985. They divided the data into twenty-year periods, measuring the per-capita annual rate of GDP growth (GROWTH) and income share of the richest twenty percent of the population (INCSHARE) at the beginning of each period. The other explanatory variables were: rate of political participation (NOFRAN), general education level (SCHOOL), level of economic development, measured as the ratio between GDP per capita and the highest level of GDP per capita in the sample in that time period (GDPGAP). The regression model of Persson and Tabellini can be summarized by the following equation:

$$GROWTH = B1 + B2\ INCSH + B3\ NOFRAN + B4\ SCHOOL + B5\ GDPGAP$$

The results of the regression are presented in the following correlation matrix (Table 1.1).

Table 1.1

Persson and Tabellini Model: Correlation Matrix

	GROWTH	GDPGAP	INCSH	SCHOOL
GDP GAP	-0.354			
INCSH	**-0.445**	-0.056		
SCHOOL	0.401	0.120	-0.713	
NOFRAN	-0.367	0.078	0.574	-0.620

The most important result is the discovery of the coefficient relating GDP growth and income share of the richest 20% to be -0.445. As Persson and

Tabellini write, "The most striking result is the effect of inequality on growth. The coefficient on INCSH is of the expected negative sign and almost always statistically significant."[8]

Persson and Tabellini provide their own explanation for their findings. They write that economies with much inequality require government programs of redistribution. These programs diminish opportunities for private accumulation and negatively affect growth.

> In a society where distributional conflict is more important, political decisions are likely to result in policies that allow less private appropriation and therefore less accumulation and growth... The main theoretical result is that income inequality is harmful for growth, because it leads to policies that do not protect property rights and do not allow full private appropriation of returns from investment.[9]

Persson and Tabellini's argument is that inequality indirectly causes inefficiencies because it prompts government policies which interfere with the market, particularly its incentives for labor and savings. This is a dramatically different theory than that of Peter and Judith Blau, which holds that inequality directly causes inefficiencies by creating psychological discontent within the society. This distinction has important ramifications: While the Blaus may be interpreted to be advocating government redistribution to eliminate inequality, Persson and Tabellini certainly cannot.

III. SYNTHESIS

Thus far, three disparate theories concerning the relationship between equality and efficiency have been presented. Basic economic theory, employed by Okun, holds that there is a negative correlation because reductions of inequality are generally effected by government tax and transfer

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[8]Persson and Tabellini p. 607.
[9]Persson and Tabellini p. 600, 617.

programs which distort economic incentives. The Blau theory holds that the correlation is positive due to the fact that inequality brings with it efficiency-harming psychological stress. The Persson-Tabellini theory holds that the correlation is positive because greater inequality leads to greater government redistribution which introduce greater inefficiencies. Is it possible to synthesize these conflicting theories?

Pre - and Post - Tax and Transfer

The conflict betweeen Okun's theory and the Persson-Tabellini theory is partially reconcilable. Both Okun and Persson / Tabellini identify redistributive programs as the prime cause of economic inefficiency. However, they conceive of different relationships between equality and redistribution. This is because while Perrson and Tabellini consider income distribution before the redistribution occasioned by taxes and transfer payments, Okun considers income distribution after redistribution. Persson and Tabellini argue that greater pre - tax / transfer inequality provides the impetus for more redistribution, less private accumulation and lowered productivity. Okun's traditional economic theory model assumes rational expectations and implicitly considers post - tax and transfer income distribution. The theory of rational expectations holds that people are aware of all factors which will affect their income in the future and take these factors into account when making economic decisions. Individuals make "the optimal use of all relevant information that is available."[10] Specifically, individuals know how much of their earnings are to be paid to the government in taxes, calculate their post-tax income, and use this figure to

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[10]William J. Baumol and Alan S. Blinder, Economics: Principles and Policy (New York: Harcourt Brace Jovanovich, 1991) p. 338.

guide their economic behavior. Since income distribution is measured after taxes and transfers, higher inequality cannot lead to more redistribution: this redistribution has already happened! In Okun's model the link is made between equality and redistribution because equality is the result of redistribution.

In all fairness, the Persson / Tabellini theory may have some relevance when considering post tax / transfer income distributions. The level of economic inequality in a particular year may have a long ranging effect on government policy, increasing government edistribution in years that follow. However, this connection is tenuous at best. The third chapter of this thesis illustrates how increasing economic inequality in the U.S. has in fact led to reductions in government redistribution.

For the purpose of this thesis I will consider equality in terms of post tax/transfer income distribution. This focuses the inquiry upon the direct effect of redistribution on individual incentives to work and save (assuming rational expectations) rather than upon the effect of pre-tax income distribution on government policy.

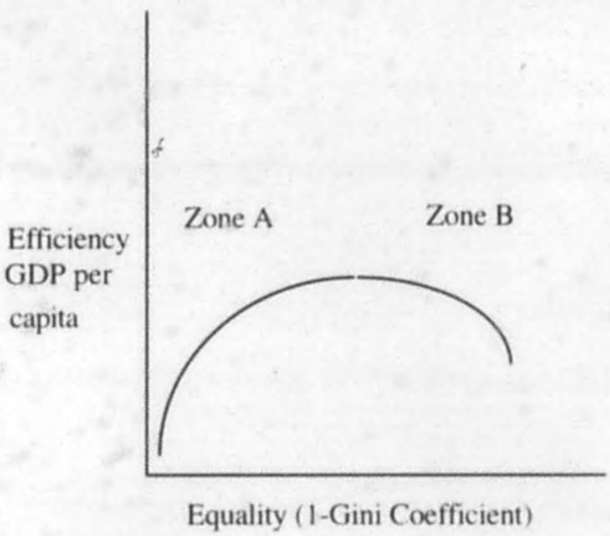
A Complex Relation

In principle, the theories of Okun and the Blaus may both be accepted as valid. Efficiency-promoting incentives and efficiency-harming social stress are both products of inequality. The net effect of inequality on efficiency is determined by which one of these two effects is predominant. It is my hypothesis that the social stress caused by inequality overwhelms the incentive effects only in circumstances of extreme inequality. Minor or moderate inequality does not produce acute and economically disruptive social stress, and therefore in circumstances of moderate inequality, the marginal effect of inequality on efficiency is positive.

This hints at a more complicated relationship between equality and efficiency than has been provided by any theory thus far. The new relationship holds that in circumstances of extreme inequality, there is a positive correlation between equality and efficiency (Zone A), yet in circumstances of moderate inequality, the correlation is negative (Zone B). This complicated relationship is the Morfit curve (Figure 1.4).

Figure 1.4

The Morfit Curve



The Morfit curve has considerable intuitive appeal. Consider what happens to economic production as income distribution becomes extremely unequal in Zone A. As resources concentrate in the hands of a smaller and smaller few, the many become impoverished. The symptoms of social stress identified by the Blaus become more pronounced. The political regime becomes progressively more unstable and the threat of revolution mounts - discouraging business investment. Consider what happens as inequality approaches its ultimate extreme. This absolute inequality (having a measured Gini coefficient of 1) is a situation in which all of a countries' income flows to one family. Such a situation is unlikely to the point of absurdity and completely unsustainable. One of two events will be triggered by levels of inequality which approach this limit - either the entire population (other than one family) will starve to death or a massive political-economic upheaval will occur. In any event, economic productivity will dip near zero. On the graph, efficiency curves asymptotically towards zero as the equality nears zero. At the other extreme, the Morfit curve terminates at the point at which the equality measure equals one (the maximum possible value of our equality measure 1-Gini Coefficient). Efficiency at this point is relatively low due to the absence of all incentive effects of inequality, but it stands well above zero since inequality does not provide the only incentive to produce (the will to survive, altruism and the joy of laboring are also strong motivators).

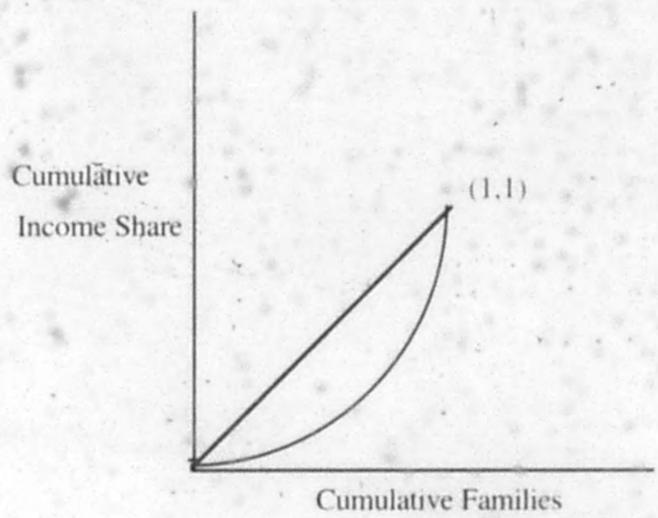
As inequality becomes more moderate, the curve enters Zone B and the correlation between equality and efficiency becomes negative for the reasons provided by Okun and basic economic theory. There is a level of inequality which maximizes efficiency, all other factors being held constant. At this optimal point, the marginal benefits of inequality are equal to its marginal costs. The incentive effects of a marginal unit of inequality upon efficiency are exactly equal to the costs in terms of social stress and political stability of that marginal unit.

Appendix to Chapter One:
The Gini Coefficient

The Gini Coefficient is a measure of inequality. It represents the divergence between a particular income distribution and a perfectly equal distribution. The Gini coefficient is most easily understood by visualizing it in graph form (Figure A.1). The graph measures cumulative income share (from 0% to 100%) on the Y axis and cumulative families, expressed as percentages of the total number of families and ranked from poorest to richest, on the X axis. The perfectly equal distribution is represented by a linear diagonal with endpoints at the origin and at the point (1,1) and with a slope of one. This line indicates that each segment of the population receives an income defined solely by the number of people it encompasses. In other words, any 5% of the population receives as much income as any other 5%. The actual distribution of income in a country most often takes the shape of a concave curve with endpoints at the origin and at the point (1,1). The concave shape indicates that a relatively richer segment of the population receives more income than a relatively poorer segment. The area between the equality line and the actual distribution curve defines the numerical value of the Gini coefficient. Its minimum possible value is zero, a situation which represents perfect equality. Its maximum possible value is one, a situation which represents perfect inequality (all income flowing to one family).

Figure A.1

Gini Coefficient



Chapter Two

The Tradeoff: An Empirical Analysis

Foreword:

On the Collection of Data

Locating and amassing statistics for empirical analysis is an extremely difficult and frustrating endeavor. This is especially the case with the project of quantifying the relationship between equality and efficiency. Statistics on income distribution are not published by any government agency and academics have had to produce these figures by analyzing government tax records. Adding to the inconvenience is the fact that few sources include data on many different countries, focusing instead on income distribution in a single country (the number of books written on the subject in the United Kingdom is astounding). The income distribution statistics used in this thesis are drawn from a broad variety of secondary sources, but the majority are taken from Charles Lewis Taylor and David A. Jodice's *World Handbook of Political and Social Indicators*. Taylor and Jodice also observe that international comparisons of income distributions are extremely rare, writing "there have been few careful collections of size distributions of income for a large number of countries. The theory of distribution has been more concerned with factors of production, in the form of wages, profits, rents, and interest, than with income by households or individuals."[11]

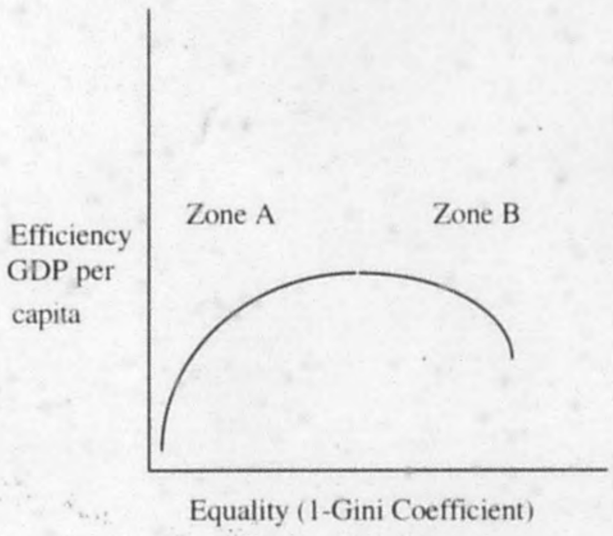
The Empirical Analysis

The purpose of this empirical analysis is to test the validity of the theory presented in Chapter One. There it is postulated that the relationship between equality and efficiency is positive in circumstances of extreme inequality and negative in circumstances of more moderate inequality. This relationship is represented graphically by the Morfit curve.

[11]Charles Lewis Taylor and David A. Jodice, *World Handbook of Political and Social Indicators*, (New Haven: Yale University Press, 1983) p. 130.

Figure 2.1

The Morfit Curve

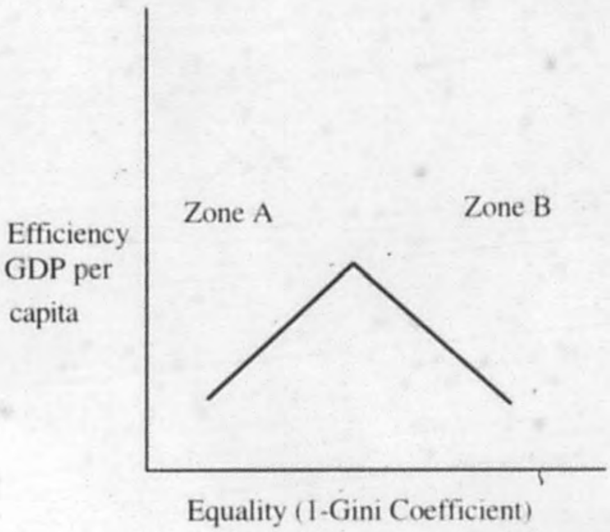


The analysis presented in this chapter is essentially a cross-sectional international comparison of data on aggregate productivity (efficiency) and equality of income distribution (equality). Linear regression analysis is used to support the theory represented by the Morfit curve. Since the Morfit curve is non-linear and convex, and the regression analysis linear, some simplification of the model is necessary. Instead of testing whether the relevant empirical data are distributed in a convex, nonlinear manner, I divide the data into two sets: set A, composed of countries with extreme inequality, and set B, composed of countries with moderate inequality. If the

Morfit curve theory is empirically valid, the former set is expected to display a positive correlation between equality and efficiency and the latter set is expected to display a negative correlation. This analysis is aimed at finding two linear relationships: a positively sloped one in Zone A and a negatively sloped one in Zone B (Figure 2.2).

Figure 2.2

Simplified Morfit Curve



The limiting factor on data collection proved to be the availability of data on income distribution. For this reason, all data used in this study is from 1970 (the year for which data on international income distribution proved to be most available). The dated nature of the data is not especially troubling because the relationship between equality and efficiency is not specific to any particular year. Since it abstracts out such topicalities as political circumstances or level of technological development, the relationship is of a timeless nature - what is found to hold in 1970 will hold in 1997, or any other year.

To measure equality, I have created an indice called Equality Ratio. This is the ratio of the proportion of national income flowing to the poorest 40% of the country to the proportion flowing to the richest 20%. A relatively high ratio indicates more equality and a relatively low ratio indicates less equality. In this study, the value of Equality Ratio ranged from a minimum of .072 (Ecuador) to a maximum of .881 (Czechoslovakia).

GDP per capita is the measure of aggregate productivity used to represent efficiency. It is expressed in 1970 U.S. dollars and is calculated using purchasing power parity. GDP per capita is used instead of the other primary measure of productivity, GDP growth, for several reasons. First, GDP growth is more strongly influenced by external variables, primarily fluctuations in the business cycle. Also, if the theory of convergence holds, rates of growth in highly developed countries are lower than those in underdeveloped countries, which distorts international comparisons.

The data are divided into two groups. Countries in which the percentage of national income flowing to the richest 20% of the population is greater than 47% are placed in group A and countries in which this figure is less than 47% are placed in group B. The dividing line between group A and group B, drawn between Uruguay and France, is arbitrary and is chosen mainly because it provides two data sets of the same size: 25 countries in each. It so happens that for the most part, the developed nations of Europe and North America are in group B and the less developed nations of Africa, Asia and South America are in group A. The legitimacy of this arbitrary division will be discussed later in the chapter.

On the following pages, the data sets for groups A and B are presented and graphed.

Table 2.1

Group A: Extreme Inequality

	top 20%	bottom 40%	Ratio bot/top	GDP p/c
Zimbabwe	69.2	8.1	0.11705202	457
Ecuador	72	5.2	0.07222222	685
Kenya	66.9	9.5	0.14200299	296
Gabon	67.5	8.5	0.12592593	844
Brazil	66.6	7	0.10510511	902
Honduras	67.8	7.3	0.10766962	434
Iraq	66.9	6.5	0.09715994	838
Senegal	62.5	9.4	0.1504	415
Colombia	59.5	10	0.16806723	843
Botswana	60.3	7.6	0.12603648	349
Tanzania	55.7	13.5	0.24236984	
Ivory Coast	58.5	10.6	0.18119658	449
Turkey	56.5	11.4	0.20176991	910
Malaysia	56.6	10.6	0.18727915	706
Costa Rica	54.8	12	0.2189781	1196
Mexico	54.4	10.3	0.18933824	1538
Venezuela	54	10.3	0.19074074	1855
Argentina	50.3	14.1	0.28031809	1995
Chile	51.4	13.4	0.26070039	1777
Hong Kong	49	15.6	0.31836735	1050
Puerto Rico	50.4	13.7	0.2718254	2077
Bahamas	50.6	12.2	0.24110672	2897
Finland	50.4	11.7	0.23214286	3075
Panama	47.4	15.2	0.32067511	1006
Italy	46.5	15.6	0.33548387	2710
Uruguay	47.5	14.2	0.29894737	1863

Figure 2.3

Group A: Extreme Inequality

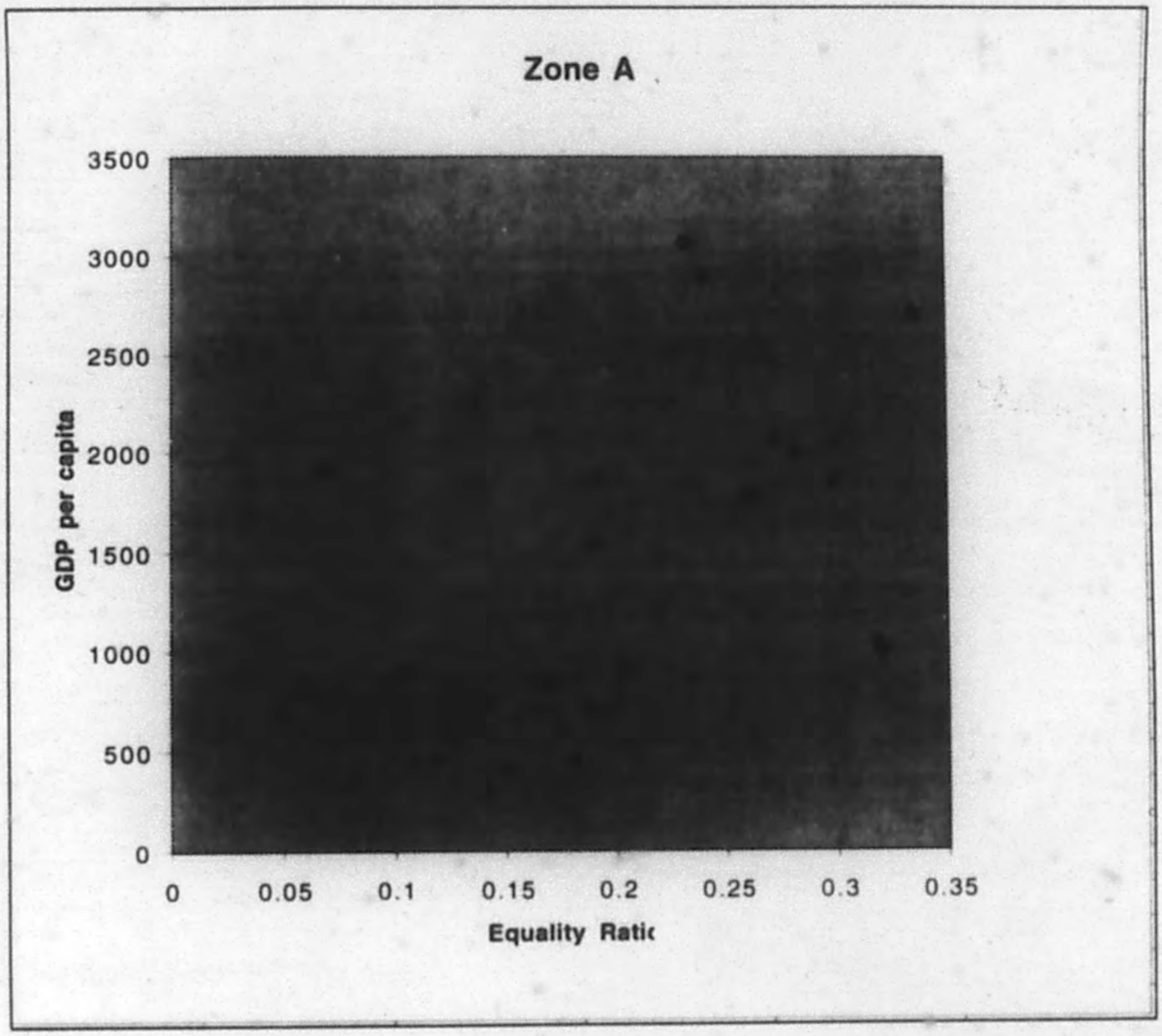


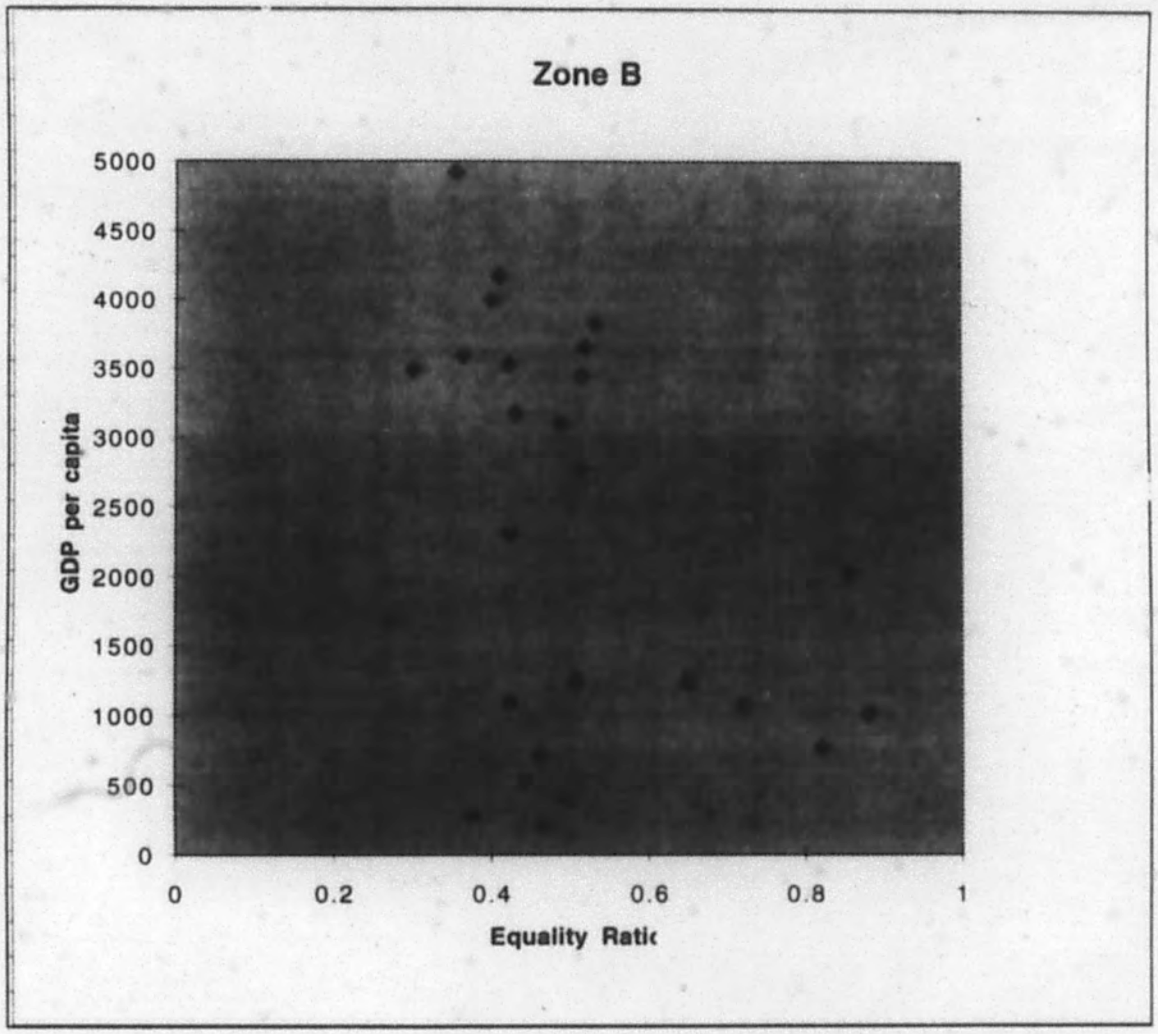
Table 2.2

Group B: Moderate Inequality

	top 20%	bottom 40%	Equality Ratio	GDP p/c
France	46.9	14.1	0.30063966	3491
W. Germany	46.2	16.8	0.36363636	3592
Barbados	44	18.6	0.42272727	1094
Sri Lanka	43.4	19.2	0.44239631	533
Netherlands	42.9	18.1	0.42191142	3528
S. Korea	45.3	16.9	0.37306843	279
Suriname	42	21.3	0.50714286	1251
Japan	41	21	0.51219512	2763
Pakistan	41.5	20.6	0.49638554	386
Bangladesh	42.3	19.6	0.46335697	223
Spain	42.2	17.8	0.42180095	2303
U.S.A.	42.8	15.2	0.35514019	4922
Denmark	42.2	16.9	0.40047393	3994
New Zealand	41.3	17.8	0.43099274	3173
Canada	41	16.8	0.4097561	4168
Australia	38.8	20.1	0.51804124	3651
U.K.	38.8	18.9	0.4871134	3097
Yugoslavia	40	18.4	0.46	714
Norway	37.3	19.2	0.51474531	3441
Sweden	37	19.7	0.53243243	3817
Poland	36.1	23.4	0.64819945	1245
Hungary	33.4	24.1	0.72155689	1075
Bulgaria	32.4	26.6	0.82098765	771
Czechoslovakia	31.1	27.4	0.88102894	1012
E. Germany	30.7	26.3	0.85667752	2019

Figure 2.4

Group B: Moderate Inequality



Regression Analysis

GDP per capita is regressed against Equality Ratio in both groups. Ordinary Least Squares estimation and tests of significance are performed using the computer program STATA. A very simple regression model is used:

GDP PER CAPITA = B1 + B2 EQUALITY RATIO

All economic experiments are imperfect. Unlike experiments in the natural sciences, the effects of a particular economic phenomenon cannot be isolated from the effects of other phenomena since no pure control groups are possible. This is especially the case with the regression model used in this study since its dependent variable is GDP per capita, a quantity which is influenced by thousands of factors. Some of these factors, most notably the amount of productive resources in a country (including human resources and technological infrastructure), are probably more significant than income distribution. No factor other than income distribution, however, is considered in this analysis. This fact should be borne in mind when considering its results.

One-tailed hypothesis testing is used in both regressions. The group A regression tests the null hypothesis that the coefficient equals zero versus the alternative hypothesis that the coefficient is greater than zero (H_0: Ratio=0 vs. H_1: Ratio>0). The group B regression tests the null against the alternative hypothesis that the coefficient is less than zero (H_0: Ratio=0 vs. H_1: Ratio<0). One-tailed testing is performed in each case because this thesis holds a bias towards discovering a positive value for the coefficient in group A and a negative value for the coefficient in group B. Tables 2.3 and 2.4 contain the results of the regressions on data set A and data set B respectively.

Table 2.3 - Group A Regression Results

	COEFFICIENT	STD. ERROR	t	P>|t|
RATIO	6833.386	1694.85	4.032	0.0005
CONSTANT	-103.7338	358.9119	-0.289	0.3875

In group A, the coefficient relating equality ratio and GDP per capita is positive (6833.386). The t value for the coefficient is 4.032, well above 1.714, the critical t value for a one tailed test at the 5% level of significance. The coefficient is significant at the 0.0005 level. R squared for the regression is 0.4141, a high value which indicates that the regression model explains the data quite well.

Table 3.4 - Group B Regression Results

	COEFFICIENT	STD. ERROR	t	P>|t|
RATIO	-3586.848	1807.297	-1.985	0.0295
CONSTANT	4092.737	963.3748	4.248	0.0

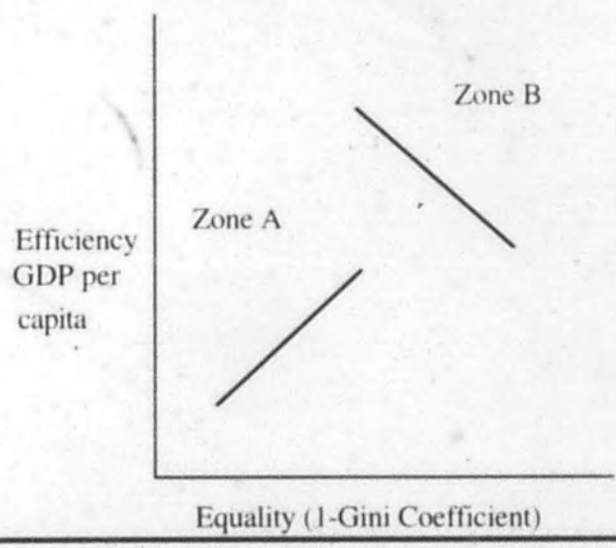
In group B, the coefficient relating equality ratio to GDP per capita is negative (-3586.848). The coefficient's t value, -1.985, has an absolute value greater than 1.714, the critical t value for a one tailed test at the 5% level of significance. The coefficient is significant at the 0.0295 level. R squared for this regression is 0.1462 - a low value which indicates that the variance of the data from the linear model is quite high. However, this is not too shocking considering the simplicity of the regression model. What is important is that the negative value of the coefficient and its significance.

The Arbitrary Division between Group A and Group B

As was stated earlier in the chapter, data is divided between groups A and B in an arbitrary manner. This unsound methodology could weaken the support that the regression analysis lends to the Morfit curve theory. The Morfit curve describes a smooth, continuous relationship between equality and efficiency. If the regression lines intersect far away from the mean value of the inequality ratio of the total data of groups A and B, making them discontinuous near the mean, then the data's support for the Morfit curve is weakened and the theory may have to be refined. This scenario is illustrated in Figure 2.5.

Figure 2.5

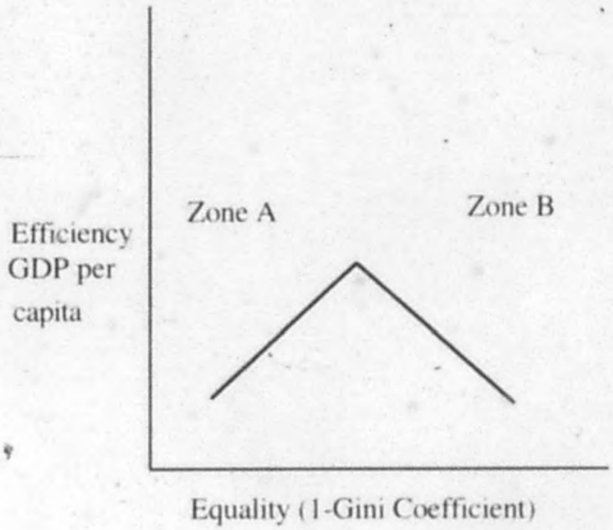
Regression lines: discontinuous near mean



If, however, the regression lines intersect near the mean value of the equality ratio, their support for the Morfit curve theory is strengthened. This situation is displayed in Figure 2.6.

Figure 2.6

Regression lines: intersect near mean



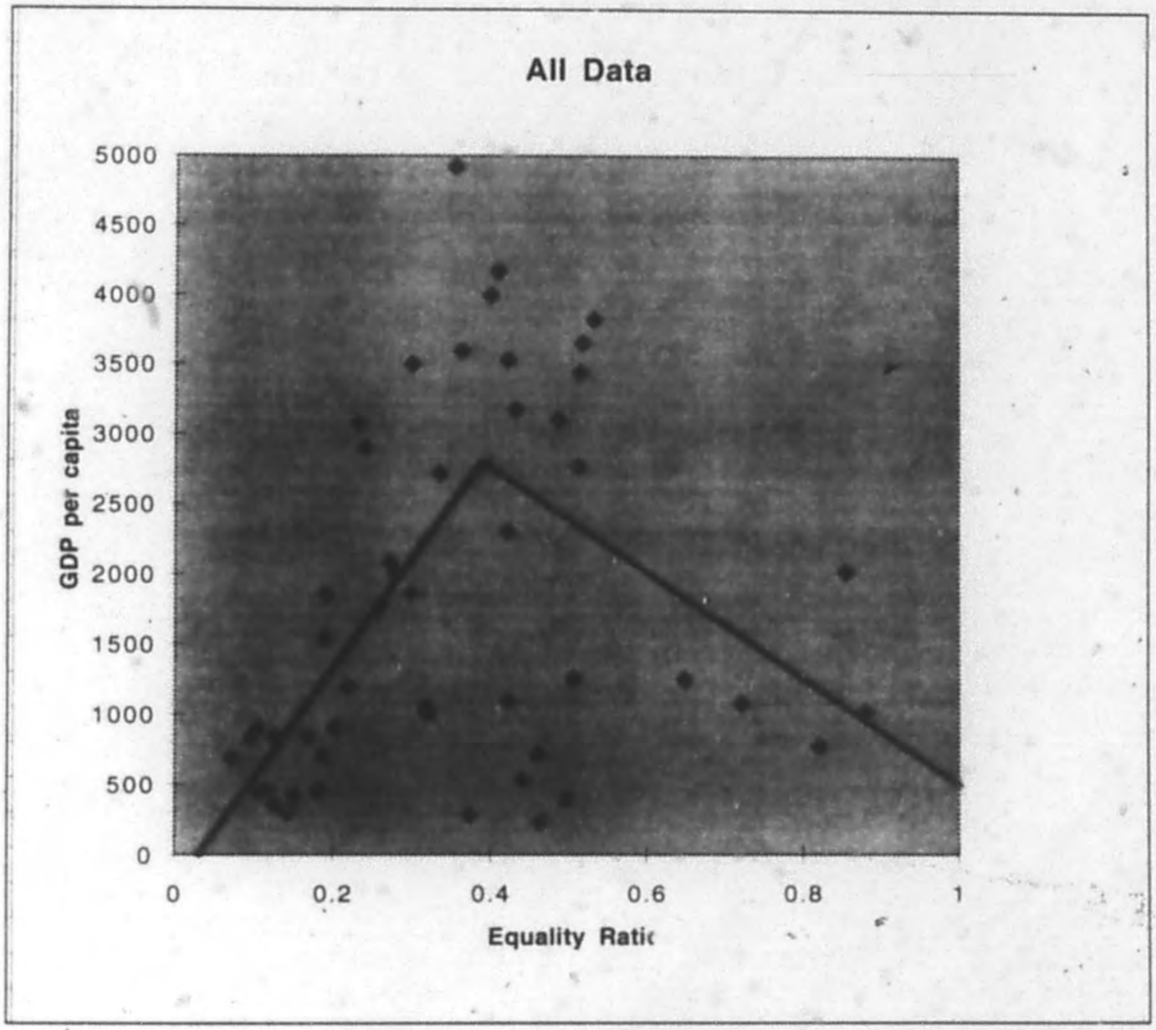
To test which of the preceding scenarios characterizes the actual data, the data for groups A and B were plotted on the same graph and then both regression lines were drawn over the scattergram. This is illustrated by figure 2.7. The two regression lines have the equations:

(Zone A) GDP PER CAPITA = -103.7338 + 6833.386 EQUALITY RATIO

(Zone B) GDP PER CAPITA = 4092.737 - 3586.848 EQUALITY RATIO

Figure 2.7

Groups A and B With Regression Lines



The regression lines intersect at the point (.4027, 2648.229). The equality ratio value at this point (.4027) is very near the mean value of .362898. Figure 2.7 is much more similar to the scenario presented in Figure 2.6 than the one presented in Figure 2.5 and supports the theory of a continuous relationship between equality and efficiency.

Conclusion

The empirical analysis presented in this chapter supports the theoretical analysis presented in Chapter One. The two regressions demonstrate a positive empirical correlation between equality and efficiency in circumstances of extreme inequality and a negative empirical correlation between equality and efficiency in circumstances of moderate inequality. As predicted by the Morfit curve, the efficiency-harming effects of social stress outweigh the efficiency-promoting incentive effects of greater inequality in situations of extreme inequality and vice versa in situations of moderate inequality. The two regression lines intersect near the mean, supporting the hypothesis of a continuous relationship between equality and efficiency.

Chapter Three

Economic Inequality in the United States

The U.S. and the Morfit Curve

The first two chapters of this thesis have posited that the marginal effect of inequality is to augment aggregate productivity in circumstances of moderate inequality and to diminish aggregate productivity in circumstances of extreme inequality. This chapter shall examine the implications of this theory for the U.S. economy.

Although the exact locus of the Morfit curve is unknown, general statements about the U.S.'s position on it can be made. Referring to the data from 1970 used in Chapter Two, the U.S. equality ratio of .355 places it very near the mean value of .362 . This implies that in 1970, the U.S. lay near the cusp of the Morfit curve, at risk of being pushed into Zone A by subsequent increases in economic inequality.

Economic Inequality in America 1970-1997

1970 marked the high point of economic equality in the United States, the end result of such government programs as Roosevelt's New Deal and Johnson's Great Society, which had been continually narrowing income disparities since 1929. The minimum wage hit its all-time peak in real terms in 1969 and in 1970 the lot of America's poor had never been better. The equalizing trend of the previous forty years, however, abruptly reversed itself in 1970. Economic inequality began to increase slowly, then continued to increase at an accelerating rate through the 1970s, 80s and 90s.

	1969	1992
Ratio: Income of Highest Quintile to Income of Lowest Quintile	**7.5**	**11**
Gini Coefficient	**3.5**	**4.0**

[12]

In 1997, economic inequality in the United States stands at its most extreme level in seventy years - since the years which saw the Roaring Twenties crash into the Great Depression. At one end of the income scale, these trends have made a great number of individuals extremely rich. In 1975, the highest-earning five percent of the population received 15% of the national income. In 1995, their share had grown to 20%.[13] From 1979 to 1995 the real earnings of this highest-earning quintile rose by 26%[14] and the real earnings of the top one percent increased by over 100%.[15] Within corporations, the earnings of executives have skyrocketed relative to those of factory workers. In 1969, chief executive officers at major corporations earned an average of forty times the pay of their workers. By 1995, this multiple had leaped to 172 and the average yearly earnings of a CEO, including salary and bonuses totaled $ 4.4 million. Today there are almost 100 American billionaires and the richest ten percent of Americans own 70% of the nation's wealth.[16]

At the other end of the income scale, increased inequality has thrown more and more people into poverty. Real annual earnings for the poorest

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[12]Samuel G. Freedman, *The End of the Line* Rolling Stone, February 20, 1997.
[13]Providence Journal-Bulletin March 9, 1997.
[14]Robert Reich, Financial Times March 3, 1997.
[15]Paul Krugman, The Age of Diminished Expectations , (Cambridge: The MIT Press, 1994) p. 25.
[16]Paul Krugman, *Rude Reality of Wealth Gap* The Baltimore Sun August 23,1995.

quintile of Americans have declined 24% since 1973 and their share of national income has fallen from 5.6% in 1975 to 4.4% in 1995.[17] Downward mobility was most acute in the 1980s, which Paul Krugman refers to as, "the first decade since the 1930s in which large numbers of Americans actually suffered a serious decline in living standards."[18] The poverty rate in the U.S. (defined as the percentage of the population earning less than 50% of the median income) increased from 15.6% in 1979 to 18.1% in 1986 (an increase of 16%).[19] Sadly, the proportion of children living in poverty hit 21.8% in 1991.[20]

Impoverishment has afflicted certain demographic groups more than others. Young people entering the workforce in the 1980s and 1990s faced a higher rates of unemployment and lower wages than their elders. The poverty rate among households headed by 20-29 year olds increased from 17.4% in 1979 to 25.2% in 1985[21] (an increase of 45%). Single-parent families have experienced poverty rates in excess of 50% during the last two decades.[22] African Americans have also been disproportionately victimized by increasing inequality. The poverty rate for Blacks in America was 49.3% in 1986, over three times the rate for Whites.

Government Programs

U.S. government social insurance programs, which have always been of a smaller scale than those of Western European countries, have been slashed in recent years - a seemingly inappropriate response to increased economic inequality. Cash transfers to the poorest 20% of Americans

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[17]Providence Journal-Bulletin March 9, 1997.
[18]Krugman p. 26.
[19]Katherine McFate et al. eds., Poverty, Inequality and the Future of Social Policy (New York: Russell Sage Foundation, 1995) p. 32.
[20]Martin Neil Bailey et. al., Growth With Equity (Washington DC: The Brookings Institute, 1993) p. 52
[21]McFate p. 33.
[22]McFate p. 34.

declined 13% from 1979 to 1989.[23] In 1979, taxes and transfers lifted 0.3% of families in poverty above the poverty line (in comparison, in the United Kingdom and France these figures were 33% and 48.2% respectively). Amazingly, by 1986 the efficacy of redistribution programs in the U.S. had dropped to a negative rate (-0.5%).[24] In 1986 taxes were pushing a greater number of people below the poverty level than transfers were lifting above it! The cutback in government redistribution efforts, however, can only account for a minute fraction of the increase in inequality. The following section examines possible explanations for the inequality explosion of the past twenty-eight years.

Theoretical Explanations

The 1969-present trend is puzzling for several reasons. Prior to 1969, economic inequality was cyclical, linked to the business cycle. Inequality rose during recessions, when the economy slowed and relatively more workers were unemployed, and fell during recoveries as unemployment fell. Policy analysis on combating inequality traditionally stressed increasing aggregate productivity, a remedy described as "a rising tide lifting all the boats." The post-1969 surge in inequality, however, has occurred independently of the business cycle: when the economy has accelerated, inequality has increased, when the economy has slowed, inequality has increased.

There is a glaring lack of consensus on the explanation for the spurt in economic inequality in the United States and the issue remains a source of heated debate between and among academics and policy makers. The root of growing inequality is surely not one single phenomenon, but rather the collective force of several. The two most often cited causes are technological

[23]Bailey p. 68.
[24]McFate p. 39.

development and the global integration of the world economy. The former explanation is championed by MIT economist Paul Krugman and the latter by his colleague Lester Thurow.

Krugman theorizes that the rapid technological innovation of the past few decades, most notably the advent of the computer, has created a "skill premium." Factory automation has replaced human laborers with machines. The demand for unskilled labor has dropped, driving wages down and forcing workers from relatively high paying manufacturing jobs into relatively low-paying jobs in the service sector. In contrast, the demand for high-skill labor has shot up, causing higher and higher wages for skilled workers. While in 1979 college-educated people earned on average 49% more than non-college educated people, now they earn 89% more.

Although technological innovation raises labor productivity and aggregate production, it's short-term effect is to throw unskilled workers into unemployment and poverty. The situation calls to mind a legendary anecdote involving Henry Ford and United Auto Workers chief Walter Reuther. Ford was proudly showing off his new assembly-line robots and said to Reuther, "These robots will never go on strike." "Yes," Reuther replied, "but how many of these robots will buy Fords?"

Lester Thurow's theory, exposited in his book The Future of Capitalism, is that the global economy is the root of America's problem of inequality. Faster and cheaper transportation and the relaxing of tariffs has left few economic barriers to prohibit American companies from relocating plants overseas. American unskilled laborers now have to compete with unskilled laborers in underdeveloped countries. The increase in labor supply has bidden wages down. Thurow is supported by the Stolper-Samuelson theory which holds that international trade will drive down the income of the factor of production in which the country does not have a comparative advantage. Since the U.S. has a comparative advantage in products which require high-skill labor and not in those which require low-skill labor, trade drives down the wages paid to low-skill U.S. workers. The fact that the share of U.S. GDP composed of manufacturing output declined from 25% in 1970 to 18.4% in 1990 also supports Thurow.[25] Krugman's arguments against Thurow are that U.S. trade with underdeveloped nations is on too small a scale (2.8% of GDP) to have significant effects on our economy, and that wage inequality has also increased in industries which are relatively immune to foreign competition. Thurow's reply is that the credible threat of U.S. firms relocating their plants overseas is enough to drive down wages.[26]

Many other phenomena have been targeted as causes of America's growing inequality. These include the decline in labor union membership - which fell from 30% of private-sector workers in 1970 to 12% in 1990.[27] Since labor unions can have dramatic effects on wages (in 1994 the average worker who belonged to a union earned 30% more than one who did not), this trend has suppressed the wages of unskilled workers.[28] The Reagan administration's economic policies of improving the living standards of the wealthy are also cited as causes of inequality. The economic growth of the 1980s was supposed to "trickle down" to the poorer classes, but the actual result was that the rich were benefited at the expense of the poor. Immigration and the increased entrance of women into the workforce are also blamed for swelling labor supply and exerting downward pressure on wages. Even the OPEC oil embargo of the 1970s, which raised energy prices,

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[25]Diane Coyle, *Economic View: Globalization and other Bugbears of Our Time* The Independent May 16, 1996.

[26]*A Tale of Two Economists* The New York Times February 16, 1997/

[27]Bailey p. 62.

[28]Rick Romell and Geeta Sharma-Jensen, *The Income Gap Riddle* Milwaukee Journal Sentinel February 6, 1996.

has been blamed for forcing the U.S. economy away from manufacturing industry and towards service.

Robert Reich, former secretary of labor under President Clinton, presents a sociological analysis of the problem. He sees growing inequality as the manifestation of a breakdown of America's implicit social compact. This compact, created in the face of the threats of the Great Depression and the second World War, was a commitment to social unity and shared prosperity. Corporate employment policy was generous, profitable companies shared their earnings with their workers through high and rising wages and provided extensive health and retirement benefits. Trade unions enjoyed large membership (35% of the private-sector workforce in the 1950s) and power. Social welfare programs provided insurance against unemployment, disability and old age. Access to a good education was guaranteed both de juro and de facto as the G.I. Bill sent millions of World War II veterans to college.[29]

Reich claims that inequality is surging because every element of the American social compact is in decline.. Profitable companies downsize, lay off workers and cut benefits. Companies are no longer guaranteeing pensions, sparking the proliferation of 401(K) savings programs, and are forcing workers to contribute to health insurance premiums.[30] The system of social insurance is reducing in scope in the face of political attack. Access to good education is unevenly distributed since schools are financed through local property taxes and the rich are abandoning public education for private schools. Reich states that this is because the United States is no longer directly threatened by its Cold War enemies and because the threat of a major

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[29]Robert Reich, Financial Times, March 3, 1997.
[30]Louis Uchitelle, *America Enters Do-It-Yourself Era* The New York Times January 13, 1997.

economic depression is not credible.[31] Without an enemy to unite against, patriotism and the sense of community suffer. The rich, confident that they will never be the recipients of government financial assistance, see no personal incentive to support social insurance. Princeton sociologist Paul Starr remarked, "What we are witnessing is the shift to taking care of yourself... the secession of the people who are the best risks and the most advantaged from the rest of the population."[32] Robert Reischauer, a senior fellow at the Brookings Institution agrees,

> When times are relatively good and the world is at peace, there is a tendency to say 'I can do it on my own.' But when the next Soviet Union arises, or a huge problem develops in how advanced medical technology can be afforded by the average person, then expect a communal response.[33]

In these times of prosperity and peace, the United States is becoming less of a community and more of a collection of atomized individuals.

From an economist's point of view, Reich's argument is less satisfactory than those of Krugman and Thurow. Although a transition in cultural mores can have major political ramifications, its influence on the behavior of corporations is questionable. It is hard to accept Reich's nostalgic claim that corporations in the past were driven less by the profit motive and more by communitarianism.

Why Inequality Causes Especially Severe Social Stress in America

The United States has several social characteristics which render it particularly vulnerable to the inequality-induced social stress discussed by Judith and Peter Blau. America's values of individualism and materialism, its political ideology and its racial diversity exacerbate the "resentment,

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[31]Robert Reich, Financial Times, March 3, 1997.
[32]Utichelle, *America Enters Do-It-Yourself Era.*
[33]Utichelle, *America Enters Do-It-Yourself Era*

frustration, hopelessness and alienation" identified by the Blaus as the bitter fruits of inequality.

Americans believe in the "American Dream," that any man, regardless of his family background and his past, can be a financial success. The American Dream is a doctrine of individualism; each individual is believed to be deserving of the power to shape his own life and responsible for the shape he gives it. Every citizen is granted the freedom to choose his or her geographic location, religion, sexual conduct, marriage partner, friends, associations, basket of consumption goods and lifestyle. He or she is expected to develop skills and to choose an occupation. As an illustration of the ethic of individualism consider the stories of self-creation and meteoric rises to financial success which permeate American literature. Horatio Alger made a career out of writing rags-to-riches novels. The Autobiography of Benjamin Franklin stresses that gains in virtue and material wealth may be acquired by all through hard work, self-discipline and education.

The ideal of individualism is empowering and inspirational, motivating people to develop their skills and to work hard. However, it also holds the individual responsible for his own condition - if he is poor it is because he has not worked hard enough. It is my opinion that this exacts a tremendous psychological toll on the poor in America, exacerbating their feelings of failure and undermining their sense of self worth. Additionally, America's doctrine of individualism keeps social insurance programs on a relatively small scale in the U.S.. Popular opinion, which holds that people should help themselves rather than be helped by the state, will not tolerate extensive government poverty relief.

America is a materialistic culture. Americans are pragmatic, rational and focused on their material well-being. Transcendental systems of thought such as Romanticism, though occasionally evident (the writings of Emerson and Thoreau for example), are rare among the populace. A society without an aristocracy, the United States is socially stratified primarily along materialistic criteria - to be a member of the upper class is to be rich. The U.S. is home to an extreme amount of interpersonal and inter familial materialistic competition and of conspicuous consumption. This is another potential source of alienation and unhappiness. In a society which values consumption so highly, those who cannot consume are marginalized.

America's democratic political system, with its universal suffrage and equal and extensive rights, recognizes the equal value of each individual. The American economy, however, with its extreme inequalities of income and wealth, stands in opposition to the fundamental values of the political system. Arthur Okun writes,

> American society proclaims the worth of every human being. All citizens are guaranteed equal justice and equal political rights... Yet at the same time, our institutions say 'find a job or go hungry,' 'succeed or suffer.'... They award prizes that allow the big winners to feed their pets better than the losers feed their children. Such is the double standard of a capitalist democracy, professing and pursuing an egalitarian political and social system and simultaneously generating gaping disparities in economic well-being.[34]

The inconsistency between America's political ideals and its economic realities may strike the poor as hypocrisy. Once a person believes that the fundamental institutions of his country are duplicitous and ungenuine he will become a cynic and will not respect the law. He will refuse to operate within a system he no longer believes in.

America's racial diversity and its history of racial conflict are catalysts for social stress. The number of African Americans in poverty has always

[34]Okun p. 1.

been disproportionate and has risen during the past three decades, nearing 50% in the mid-1980s. Regardless of how much of this is due to racial discrimination, Blacks see their economic misfortunes as products of racism. Believing that they are competing in an unfair system, Black Americans are particularly vulnerable to the alienation, resentment and anarchism which will harm the U.S.'s productivity.

Elevated Social Stress in America

Indicators of social stress in America have risen coincidentally with inequality over the past thirty years. Violent crime rates doubled during the 1980s and in each of the past fifteen years over 20,000 Americans have been murdered.[35] Disturbingly, juvenile violence has risen even faster. The rate of murder committed by children aged 14 to 17 increased 22% from 1990 to 1994.[36] 5,721 children died of gunfire in 1993, an average of over sixteen per day and nearly twice as many as in 1983.[37]

Crime exacts many economic costs in addition to lost lives. A Brookings Institution paper states,

> As a practical matter [crime] acts as a drag on our economy. The prevalence of serious crime not only requires heavy public expenditure to capture, try and imprison lawbreakers, it also encourages wasteful private spending to protect American homes and business.[38]

These costs are conspicuous. State expenditures on law enforcement and incarceration have been forced to rise to deal with the rise in crime while Robert Dahl and others have observed the proliferation of "gated communities" in America, wealthy neighborhoods which hire private

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[35] Ted Robert Gurr, *The Nation is Caught in a Third Great Crime Wave* St. Petersburg Times April 15, 1989.
[36] Rebecca Carr, *Youth Crime Surge Termed a Nightmare* The Stuart News October 28, 1996.
[37] Susan Cornwell, *Sixteen U.S. Kids Die of Gunfire Each Day* Reuters North American Wire March 12, 1997.
[38] Bailey p. 5.

security forces and erect physical barriers between themselves and the outside world. These costs will continue to mount so long as inequality continues to increase.

The 1970s, 80s and 90s witnessed the disintegration of the American family. The divorce rate increased from 9.4 per thousand women in 1960 to 21.2 per thousand in 1986. Over the same time period, the percentage of births out of wedlock increased from 5.3% to 23.4%.[39] In 1990 one out of every six American families was headed by an unmarried woman (up from one in nine in 1970). Among the poorest 10% of Americans, the percentage of families headed by a single woman rose from 37% to 67% between 1970 and 1990. Over 8,000 accounts of child abuse were reported in 1995, an increase of 25% from the 1990 figure.

The full cost of these demographic phenomena has not yet been accrued. A generation of millions of fatherless children is just beginning to enter the labor market. Their aggregate psychological discontent is unmeasurable. Their ambition, trainability and employability remains to be seen.

The U.S. and the Morfit Curve (1997)
or To the Future - the Theory

The 1969-1997 inequality boom may have pushed the U.S. over the cusp of the Morfit curve and into Zone A, where the marginal effect of inequality on aggregate productivity is negative. In 1997, inequality may be bringing about a level of efficiency-harming social stress which outweighs its efficiency-promoting incentive effects.

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[39] McFate p. 11.

This has some interesting implications for U.S. economic policy. The most potent defense of economic conservatism is the appeal to U.S. economic productivity. This argument, based on a theory very similar to Arthur Okun's theory discussed in Chapter One, holds that redistributive policies are undesirable because they subvert aggregate economic productivity. This thesis, however, holds that just the opposite may be the case: redistributing income from the rich to the poor may actually boost the U.S.'s economic performance. If this is the case, most objections to the welfare state in America are nullified and the scope of redistribution programs should be expanded.

To the Future - the Reality

Social insurance programs have been a major political issue in the past year and the welfare state in the United States is in the process of being demolished. The welfare bill of 1996, authored by Republican legislators and signed by President Clinton in August 1996, is notable as marking the official termination of the federal government's guarantee to provide cash assistance for the poor. In the future, states will receive block grants of federal funds and will have the power to administer their own welfare programs as they see fit. The bill also requires benefit recipients to find employment within two years and sets a five year lifetime limit on benefits. Immigrants arriving after August 1996 will be ineligible for most federal benefits. The cases of non-citizens will be reviewed and most of them will be denied Medicaid and other government services. Supplementary income for disabled children will be cut. The federal food stamps program shall be reduced in scope. The federal guarantee of cash aid for poor children under Aid for Families with Dependent Children (AFDC) will be revoked. The bill states that "No individual or family shall be entitled to any benefits or services" under AFDC.[40]

President Clinton defended the welfare reform bill by stating that generous social insurance is a disincentive for work and that the reductions in benefits will actually help the poor by forcing them to take jobs.

> A long time ago I concluded that the current welfare system undermines the base values of work, responsibility and family, trapping generation after generation in a dependency and hurting the very people it was designed to help. Today we have a historic opportunity to make welfare what it was meant to be, a second chance not a way of life... First and foremost, it should be about moving people from welfare to work. It should impose time limits on welfare... the best anti-poverty program is still a job.[41]

Later the President boasted that, "There are 1.3 million fewer people on welfare today than when I took office."[42] Although the implication was that this figure represents people finding work and lifting themselves out of poverty, it is due more to increasingly restrictive eligibility requirements for benefits.

Clinton's endorsement of the Republican welfare reform has been criticized for being part of his 1996 re-election bid rather than a carefully thought out policy move. NPR commentator Daniel Schorr bitterly remarked on the radio that

> This drastic piece of legislation is dressed up in the language of tough love. Speaker Newt Gingrich says that what is wrong with the present system is not that people abuse welfare but that welfare abuses people. Soon we shall see who is abusing whom. Ending the 60-year-old federal entitlement means taking a plunge into the unknown without much of a safety net to catch the children who are inevitably the most vulnerable."[43]

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[40]Robert Pear, *Overhauling Welfare: A Look at the Year Ahead* The New York Times August 7, 1996.
[41]President William J. Clinton, *Statement on the Welfare Reform Bill* July 31, 1996.
[42]Clinton, *Statement on the Welfare Reform Bill* July 31, 1996.
[43]Daniel Schorr, *News Analysis* All Things Considered National Public Radio, August 1, 1996.

Whatever the rhetoric may be, it is indisputable that the welfare reform bill, by cutting government benefits for the poor, will exacerbate the problem of economic inequality in America.

Economists agree that to combat rising inequality, the scope of government aid to the poor should be enlarged, not reduced. In particularly, education is widely regarded as the most effective weapon for reducing inequality in the long run. The U.S. economy has undergone structural changes which have increased the demand for skilled labor and reduced the demand for unskilled labor. In order to match this demand, the government must provide training in basic skills and make higher education affordable to all. The U.S. economy of the future will continue to be dynamic, buffeted by external shocks and occasionally undergoing structural transformations. In order to move workers from declining industries to growing ones, education should be flexible and directed towards providing the skills needed by the growing sectors of the economy.

A Final Word

Considering the American twenty-eight year trend of increasing economic inequality and given the U.S. government's negative attitude towards social insurance, there is no reason to believe that inequality will not continue to increase. As the poor become progressively disenchanted, orphaned and violent and the rich isolate themselves behind more and more walls, alarms and security guards, the output of the U.S. economy will fall farther and farther behind its potential.

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This thesis represents my own work in accordance with University regulations